

- 6** Barium dithionate, $\text{BaS}_2\text{O}_6 \cdot 2\text{H}_2\text{O}$, is soluble in water.

$\text{S}_2\text{O}_6^{2-}$ ions slowly decompose in acidic solution.



3.513 g of $\text{BaS}_2\text{O}_6 \cdot 2\text{H}_2\text{O}$ is dissolved in some water and the solution made up to the mark with $\text{HCl}(\text{aq})$ in a 100 cm^3 volumetric flask.

At time x min, a white precipitate of mass 0.661 g is present in the flask.

What is the concentration of BaS_2O_6 in the volumetric flask at time x min?

[A_r : Ba, 137.3; S, 32.1; O, 16.0; H, 1.0]

- A** $0.0077 \text{ mol dm}^{-3}$
- B** $0.0090 \text{ mol dm}^{-3}$
- C** $0.077 \text{ mol dm}^{-3}$
- D** $0.090 \text{ mol dm}^{-3}$